

Supplementary Material

Testing Molecular Specificity in Electrophysiology-to-Transcript Prediction with Matched Negative Controls

ICCKE 2026

Scope. This supplement reports the complete donor-grouped out-of-fold results and the validated primary and sensitivity analyses supporting the main text. Tables S1–S50 are provided as machine-readable CSV files and Figures S1–S5 in publication-ready formats. All entries are taken directly from validated analysis outputs; no model was refitted and no numerical result was recalculated. Complete CSV tables are supplied where only a concise display is possible within the page limit.

S1 Data Sources and Cohort Construction

Cells were joined one-to-one by `cell_specimen_id`. The fixed analytic cohort contains 3,410 mouse visual-cortex inhibitory neurons from 871 donor groups and 6 broad subclasses. Tables S1–S3 provide the cohort characteristics, donor counts, and canonical crosswalk. Donor identity defined every outer split. Raw Allen data are not redistributed.

Cohort composition, donor counts, and cell-level crosswalk (Tables S1–S3).

Item	Value
Paired MAT cells before public-metadata matching	3,411
Unmatched specimen excluded	1
Cells in analytic cohort	3,410
Donor groups	871
Broad subclasses	6
Outer donor-disjoint folds	3 (1,137 / 1,136 / 1,137 cells)
Canonical cell-crosswalk rows	3,410
Modeled electrophysiology predictors	23

S2 Electrophysiology Predictors

The source matrix contained 24 variables. Modeling retained `rheobase_i` and removed its exact duplicate, `stimulus_amplitude_0_long_square`, leaving 23 nonredundant IPFX predictors; 44 waveform principal components were excluded. Release-level standardization and clipping at ± 6 SD occurred upstream. Physical-unit inversion from the saved affine transform is approximate, and observations at the clipping boundaries remain censored. Only downstream median imputation and model-specific scaling were fitted within each outer-training fold.

Electrophysiological predictors used in the 23-feature analysis (Table S4).

No.	Exact feature key	Unit	No.	Exact feature key	Unit
1	ap_1_threshold_v_short_square	mV	13	sag_measured_at	mV
2	ap_1_peak_v_short_square	mV	14	rheobase_i	pA
3	ap_1_upstroke_short_square	V/s (equivalent mV/ms)	15	ap_1_threshold_v_0_long_square	mV
4	ap_1_downstroke_short_square	V/s (equivalent mV/ms)	16	ap_1_peak_v_0_long_square	mV
5	ap_1_upstroke_downstroke_ratio_short_square	dimensionless	17	ap_1_upstroke_0_long_square	V/s (equivalent mV/ms)
6	ap_1_width_short_square	s	18	ap_1_downstroke_0_long_square	V/s (equivalent mV/ms)
7	ap_1_fast_trough_v_short_square	mV	19	ap_1_upstroke_downstroke_ratio_0_long_square	dimensionless
8	short_square_current	pA	20	ap_1_width_0_long_square	s
9	input_resistance	MΩ	21	ap_1_fast_trough_v_0_long_square	mV
10	tau	s	22	avg_rate_0_long_square	Hz
11	v_baseline	mV	23	latency_0_long_square	s
12	sag_nearest_minus_100	dimensionless			

S3 Molecular Module Definitions

Targets were unweighted means of $\log_2(\text{CPM} + 1)$ across 6 prespecified gene modules; CPM denominators used all 45,768 count rows. Exact-symbol coverage was 95/96. Exact **Scn2a** was absent and was not replaced by **Scn2a1**; the primary NaV score therefore contains 6 observed genes. NaV7 is reported only as a separate sensitivity.

Gene-module definitions and exact-symbol coverage (Tables S5–S6).

Module	Used/requested	Genes used	Missing exact symbol
NaV	6/7	Scn1a, Scn3a, Scn8a, Scn9a, Scn10a, Scn11a	Scn2a
Kv	40/40	Kcna1, Kcna2, Kcna3, Kcna4, Kcna5, Kcna6, Kcna7, Kcna10, Kcnb1, Kcnb2, Kcnc1, Kcnc2, Kcnc3, Kcnc4, Kcnd1, Kcnd2, Kcnd3, Kcnf1, Keng1, Keng2, Keng3, Keng4, Kcnq1, Kcnq2, Kcnq3, Kcnq4, Kcnq5, Kens1, Kens2, Kens3, Kcnv1, Kcnv2, Kenh1, Kenh2, Kenh3, Kenh4, Kenh5, Kenh6, Kenh7, Kenh8	–
CaV	10/10	Cacna1a, Cacna1c, Cacna1d, Cacna1f, Cacna1a, Cacna1b, Cacna1e, Cacna1g, Cacna1h, Cacna1i	–
HCN	4/4	Hcn1, Hcn2, Hcn3, Hcn4	–
GABAA	19/19	Gabra1, Gabra2, Gabra3, Gabra4, Gabra5, Gabra6, Gabrb1, Gabrb2, Gabrb3, Gabrg1, Gabrg2, Gabrg3, Gabrd, Gabre, Gabrp, Gabrq, Gabrr1, Gabrr2, Gabrr3	–
iGluR	16/16	Gria1, Gria2, Gria3, Gria4, Grik1, Grik2, Grik3, Grik4, Grik5, Grin1, Grin2a, Grin2b, Grin2c, Grin2d, Grin3a, Grin3b	–

S4 Model Specifications

Elastic Net, XGBoost, and a small multilayer perceptron (MLP) were evaluated on the same donor-disjoint outer folds. Candidate selection used one 20% donor-disjoint inner split within each outer-training set. Tables S7–S9 provide the fixed search spaces, MLP training and early-stopping procedure, and every fold-specific selected setting.

Model search spaces, training procedures, and selected hyperparameters (Tables S7–S9).

Model	Candidate-specific values	Training and preprocessing procedure
Elastic Net	$\alpha \in \{.001, .010, .100, 1.000\}$, ℓ_1 ratio $\in \{.1, .5, .9\}$ (12 candidates)	Maximum 5,000 iterations; seed 42; fold-local median imputation and standardization.
XGBoost	(250/2/.05/1); (400/2/.03/3); (250/3/.05/3); (300/4/.03/5) (trees/depth/rate/ L_2)	Subsample .8; column sample .8; min-child weight 1; $L_1 = 0$; gamma 0; objective <code>reg:squarederror</code> ; tree method <code>hist</code> ; 1 job; seed 42; fold-local median imputation.
MLP	Hidden layers 64/32; activation relu; solver adam; $L_2 = .001$; <code>learning_rate=constant</code> ; <code>learning_rate_init=.001</code> ; <code>batch_size=auto</code> ; <code>shuffle=True</code> ; seeds 42/123/2026	<code>inner_split_method=GroupShuffleSplit</code> ; <code>inner_validation_fraction=.2</code> ; <code>inner_validation_grouped=True</code> ; <code>early_stopping=group_aware_manual</code> ; <code>sklearn_early_stopping=False</code> ; estimator validation fraction 0.15 is inert. Maximum 300 epochs; patience 20; <code>min_delta = 1e - 05</code> ; epoch selected by minimum validation RMSE; fresh full outer-train refit; <code>n_iter_=selected_epoch</code> for every saved model.

S5 Complete Primary Performance

Pooled donor-grouped OOF R^2 was the primary predictive metric. Table S10 reports every model–module combination with R^2 , MAE, RMSE, and Spearman ρ ; Table S11 gives the corresponding fold-level results. Individual-model confidence intervals are available only for the 24-feature reference analysis (Table S12) and are not presented for the 23-feature analysis. Table S13 gives the direct 23-feature-versus-24-feature sensitivity.

Complete pooled OOF performance for the 23-feature analysis (Table S10; $n = 3,410$).

Module	Model	R^2	MAE	RMSE	Spearman ρ
NaV	Dummy	-.003	.670	.876	-.043
NaV	Elastic Net	.121	.633	.820	.355
NaV	XGBoost	.133	.626	.814	.380
NaV	MLP	.082	.646	.838	.346
NaV	Subclass Ridge	.102	.633	.829	.300
NaV	Ephys + subclass Ridge	.136	.626	.813	.388
Kv	Dummy	-.002	.423	.553	-.041
Kv	Elastic Net	.046	.413	.539	.221
Kv	XGBoost	.046	.410	.539	.226
Kv	MLP	-.015	.426	.556	.219
Kv	Subclass Ridge	.042	.413	.541	.179
Kv	Ephys + subclass Ridge	.065	.408	.534	.262
CaV	Dummy	-.002	.734	.965	-.035
CaV	Elastic Net	.094	.674	.918	.361
CaV	XGBoost	.106	.663	.912	.395
CaV	MLP	.035	.696	.947	.342
CaV	Subclass Ridge	.104	.664	.913	.351
CaV	Ephys + subclass Ridge	.114	.662	.908	.399
HCN	Dummy	-.001	.932	1.172	-.023
HCN	Elastic Net	.064	.904	1.133	.256
HCN	XGBoost	.057	.906	1.138	.248
HCN	MLP	.011	.926	1.165	.221
HCN	Subclass Ridge	.041	.910	1.147	.171
HCN	Ephys + subclass Ridge	.057	.907	1.138	.248
GABAA	Dummy	-.001	.634	.794	-.032
GABAA	Elastic Net	.260	.526	.682	.528
GABAA	XGBoost	.329	.494	.650	.574
GABAA	MLP	.314	.501	.657	.567

Table S10 continued

Module	Model	R^2	MAE	RMSE	Spearman ρ
GABAA	Subclass Ridge	.177	.551	.720	.391
GABAA	Ephys + subclass Ridge	.319	.499	.655	.574
iGluR	Dummy	-.002	.765	1.000	-.019
iGluR	Elastic Net	.258	.612	.861	.602
iGluR	XGBoost	.295	.581	.840	.631
iGluR	MLP	.206	.622	.891	.582
iGluR	Subclass Ridge	.327	.560	.820	.637
iGluR	Ephys + subclass Ridge	.340	.557	.812	.667

Change in R^2 : 23-feature minus 24-feature reference analysis (Table S13).

Module	Elastic Net	XGBoost	MLP
NaV	-.000096	.003851	-.010114
Kv	-.000003	-.003146	-.019469
CaV	.000081	.000306	-.013158
HCN	-.000005	-.003854	.000731
GABAA	.000001	-.000816	-.011799
iGluR	-.000013	.000682	-.019147

S6 Formal Model Comparisons

The prespecified contrasts used 5,000 paired donor-cluster bootstrap draws. Positive ΔR^2 favors the first model named. Table S14 contains the complete metric-specific comparisons; Table S15 gives the prespecified R^2 family with Benjamini–Hochberg adjustment. Corrected positive support for XGBoost over Elastic Net was confined to GABAA and iGluR. Positive values favor the first model; a negative BH-resolved contrast indicates that the first model performed worse.

Paired donor-cluster bootstrap comparisons for pooled OOF R^2 (Table S15).

Module	Contrast	ΔR^2	95% CI	BH q	BH-resolved
NaV	XGB - EN	.013	[-.001, .026]	.0825	no
NaV	MLP - EN	-.038	[-.061, -.017]	.0011	yes
NaV	MLP - XGB	-.051	[-.071, -.032]	.0006	yes
Kv	XGB - EN	-.0001	[-.015, .015]	.9942	no
Kv	MLP - EN	-.061	[-.090, -.033]	.0006	yes
Kv	MLP - XGB	-.061	[-.087, -.037]	.0006	yes
CaV	XGB - EN	.012	[-.001, .025]	.0825	no
CaV	MLP - EN	-.059	[-.084, -.035]	.0006	yes
CaV	MLP - XGB	-.071	[-.095, -.048]	.0006	yes
HCN	XGB - EN	-.007	[-.018, .004]	.2189	no
HCN	MLP - EN	-.053	[-.073, -.033]	.0006	yes
HCN	MLP - XGB	-.046	[-.066, -.027]	.0006	yes
GABAA	XGB - EN	.069	[.052, .086]	.0006	yes
GABAA	MLP - EN	.054	[.030, .077]	.0006	yes
GABAA	MLP - XGB	-.015	[-.037, .006]	.1818	no
iGluR	XGB - EN	.036	[.021, .053]	.0006	yes
iGluR	MLP - EN	-.052	[-.079, -.028]	.0006	yes
iGluR	MLP - XGB	-.088	[-.117, -.064]	.0006	yes

S7 Permutation Diagnostics

Both permutation families were applied to saved OOF predictions without model refitting; they are association diagnostics rather than algorithm-level randomization tests. The cell-level family permuted truth within subclass. The donor-level family permuted subclass-adjusted donor truth summaries across 871 donors.

Permutation diagnostics based on saved OOF predictions, without model refitting (Table S16; 5,000 permutations).

Module	Model	Cell Spearman ρ	Cell p	Donor adjusted Spearman ρ	Donor p
NaV	Elastic Net	.355	.0002	.241	.0002
NaV	XGBoost	.380	.0002	.238	.0002
NaV	MLP	.346	.0002	.245	.0002
Kv	Elastic Net	.221	.0002	.138	.0004
Kv	XGBoost	.226	.0002	.180	.0002
Kv	MLP	.219	.0002	.192	.0002
CaV	Elastic Net	.361	.0002	.092	.0066
CaV	XGBoost	.395	.0002	.133	.0002
CaV	MLP	.342	.0002	.142	.0002
HCN	Elastic Net	.256	.0002	.101	.0040
HCN	XGBoost	.248	.0002	.116	.0004
HCN	MLP	.221	.0002	.151	.0002
GABAA	Elastic Net	.528	.0002	.329	.0002
GABAA	XGBoost	.574	.0002	.380	.0002
GABAA	MLP	.567	.0002	.386	.0002
iGluR	Elastic Net	.602	.0002	.143	.0004
iGluR	XGBoost	.631	.0002	.215	.0002
iGluR	MLP	.582	.0002	.204	.0002

S8 Matched Negative Controls

Each observed module was compared with 1,000 same-size gene sets matched on gene-level mean $\log_2(\text{CPM}+1)$ and detection rate; matching did not directly use co-expression or within-module coherence. These two quantities were standardized over eligible non-module genes. For each target gene, up to 128 nearest candidates were ordered by standardized Euclidean distance and selected without replacement; all primary-module genes were excluded. A set was accepted only when maximum and mean per-gene distances were at most .50 and .25, respectively, and absolute module-level expression and detection imbalances were at most .10 and .10; up to 200 deterministic proposals were allowed. The observed-module XGBoost configuration was reused without control-outcome tuning. All 6,000 sets met these criteria. The percentile is $100 \Pr(R_{control}^2 \leq R_{observed}^2)$ and the empirical p value is $(1 + \#\{R_{control}^2 \geq R_{observed}^2\})/(B + 1)$.

Primary matched-control specificity analysis on unadjusted module scores (Table S18; separate 6-test family).

Module	Observed R^2	Null median	Null P95	Percentile	p	BH q	BH-supported
NaV	.133	.029	.131	95.1	.0500	.0999	no
Kv	.046	.101	.185	10.1	.8991	.8991	no
CaV	.106	.056	.163	80.5	.1958	.2937	no
HCN	.057	.023	.123	74.2	.2587	.3105	no
GABAA	.329	.071	.200	99.8	.0030	.0180	yes
iGluR	.295	.084	.230	98.7	.0140	.0420	yes

Matching diagnostics and fixed-versus-retuned sensitivity (Tables S19 and S21). Retuning used 50 controls per module; ΔR^2 is retuned minus fixed.

Module	Passed	Mean dist.	Max dist.	Max $ \Delta z_E $	Max $ \Delta z_D $	Rejected proposals	Median ΔR^2
NaV	1,000/1,000	.043	.174	.052	.042	0	.0007
Kv	1,000/1,000	.072	.328	.032	.038	0	.0034
CaV	1,000/1,000	.068	.211	.058	.057	0	.0016
HCN	1,000/1,000	.071	.169	.081	.082	0	.0038
GABAA	1,000/1,000	.070	.271	.047	.068	0	.0079
iGluR	1,000/1,000	.156	.497	.099	.082	56,007	.0039

Note. Rejected proposals are unsuccessful deterministic candidate-set proposals generated before obtaining the 1,000 accepted matched sets; they are not excluded cells or failed final controls.

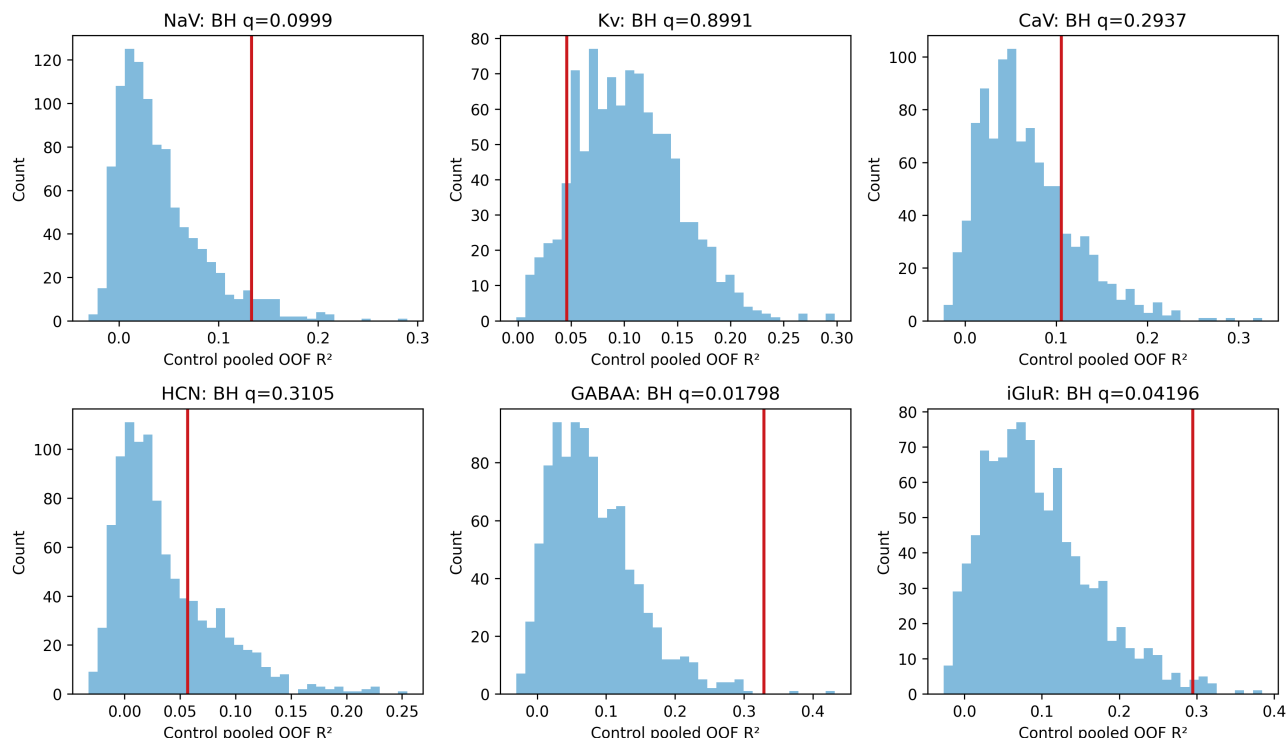


Figure S1. Matched negative-control specificity. The primary raw family is kept separate from the technical-residualized sensitivity family.

S9 Technical Targets and Residualized Specificity

Detected-gene count was predictable, whereas total library count was not above the mean baseline. Fold-local technical residualization changes both target and null distribution. The residualized results are therefore a separate six-test sensitivity family and are never pooled with the primary raw family.

Technical-target OOF performance (Table S22).

Target	Model	R^2	MAE	RMSE	ρ
$\log_{10}(\text{genes detected})$	Elastic Net	.140	.058	.080	.408
$\log_{10}(\text{genes detected})$	XGBoost	.135	.057	.080	.417
$\log_{10}(\text{total counts})$	Elastic Net	-.001	.097	.166	-.024
$\log_{10}(\text{total counts})$	XGBoost	-.019	.099	.167	.060

Matched-control specificity after fold-local technical residualization (Table S23; separate 6-test family).

Module	Observed R^2	Null median	Null P95	Percentile	p	BH q	BH-supported
NaV	.113	.002	.097	96.8	.0330	.0495	yes
Kv	.041	.021	.090	72.6	.2747	.2747	no
CaV	.175	.015	.152	96.8	.0330	.0495	yes
HCN	.043	.005	.115	80.3	.1978	.2374	no
GABAA	.383	.020	.135	100.0	.0010	.0060	yes
iGluR	.267	.057	.198	99.0	.0110	.0330	yes

S10 Module Internal Consistency and Control Coherence

Balanced split-half analysis measures internal consistency; it is not test–retest reliability and not a formal noise ceiling. Tables S24–S28 keep the high-resolution observed partition analysis separate from the control-comparable analysis. Reliability-contextualized ratios are descriptive and should not be read as corrected prediction accuracy.

Observed and matched-control split-half consistency, with descriptive reliability-contextualized R^2 ratios (Tables S27–S28).

Target	Obs. split-half SB	Control median SB	SB pct.	R^2	R^2/SB	Ratio pct.
NaV	.207	.126	86.5	.133	.643	83.3
Kv	.537	.529	57.0	.046	.085	7.9
CaV	.521	.232	100.0	.106	.203	38.6
HCN	.207	.111	91.5	.057	.274	56.4
GABAA	.587	.336	99.9	.329	.561	96.6
iGluR	.641	.332	100.0	.295	.460	84.3
NaV7	.274	.149	92.8	.078	.283	54.0

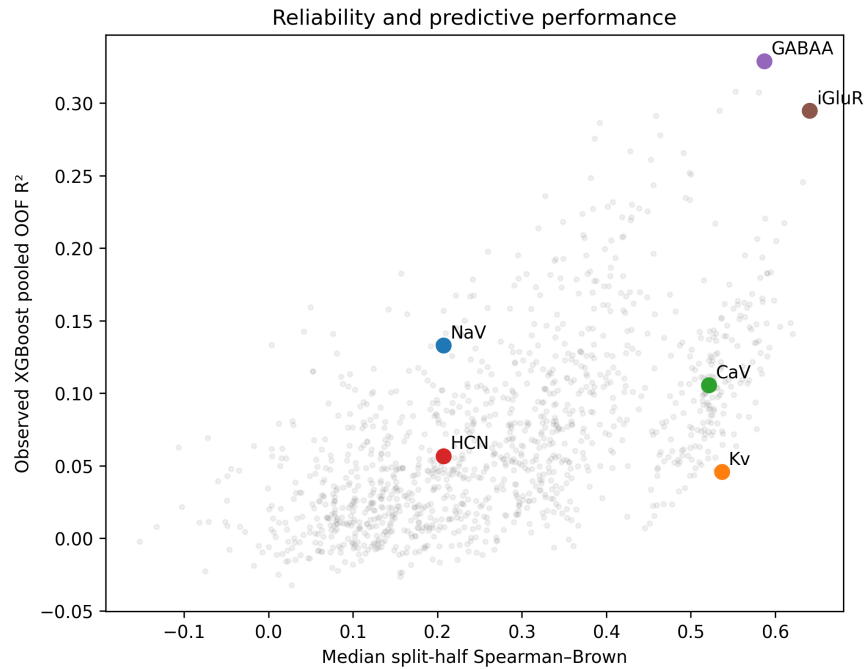


Figure S2. Observed module consistency in the context of matched-control consistency. SB denotes Spearman–Brown correction.

S11 NaV and Gene-Composition Sensitivities

NaV6 is the prespecified primary target with 6 observed genes. NaV7 is a distinct 7-gene sensitivity using the exact release symbol **Scn2a1**; it does not replace missing **Scn2a** and is outside the primary matched-control BH family. Tables S29–S31 give the complete NaV6/NaV7 definitions and model metrics. Target-definition (Tables S32–S33) and leave-one-gene-out (LOGO; Table S34) results are available only under the 24-feature reference specification; no 23-feature refit was performed. Figure S5 summarizes the NaV comparison.

Prespecified NaV6 target and separate NaV7 sensitivity (Table S29).

Target	Genes	R^2	Pct.	p	BH q	Median SB	Role
NaV6	6	.133	95.1	.0500	.0999	.207	primary
NaV7 (+Scn2a1)	7	.078	79.1	.2098	–	.274	separate sensitivity

Target-definition sensitivity under the 24-feature reference specification (Table S33; XGBoost).

Module	Mean R^2	Std. mean R^2	PC1 R^2	Max $ \Delta R^2 $	PC1 variance	Class
NaV	.129	.094	.207	.078	.271	moderately sensitive
Kv	.049	.049	.291	.242	.091	composition-sensitive
CaV	.105	.050	.060	.056	.278	moderately sensitive
HCN	.061	.039	.087	.026	.316	robust
GABAA	.330	.215	.503	.173	.169	composition-sensitive
iGluR	.294	.186	.221	.108	.279	composition-sensitive

Leave-one-gene-out sensitivity under the 24-feature reference specification (Table S34).

Module	Omitted gene	Model	R^2	ΔR^2	Classification
NaV	Scn3a	Dummy	-.002	.001	baseline reference
NaV	Scn3a	Elastic Net	.293	.172	stronger after omission
NaV	Scn3a	XGBoost	.307	.178	stronger after omission
NaV	Scn3a	Subclass Ridge	.278	.177	stronger after omission
NaV	Scn9a	Dummy	-.002	.001	baseline reference
NaV	Scn9a	Elastic Net	.013	-.107	weakened
NaV	Scn9a	XGBoost	-.010	-.139	weakened
NaV	Scn9a	Subclass Ridge	.007	-.095	weakened
HCN	Hcn2	Dummy	-.001	-.0003	baseline reference
HCN	Hcn2	Elastic Net	.034	-.030	mixed or partial
HCN	Hcn2	XGBoost	.032	-.028	mixed or partial
HCN	Hcn2	Subclass Ridge	.010	-.031	mixed or partial
HCN	Hcn3	Dummy	-.001	-.0001	baseline reference
HCN	Hcn3	Elastic Net	.123	.059	stronger after omission
HCN	Hcn3	XGBoost	.131	.071	stronger after omission
HCN	Hcn3	Subclass Ridge	.081	.040	mixed or partial

S12 Subclass Analyses

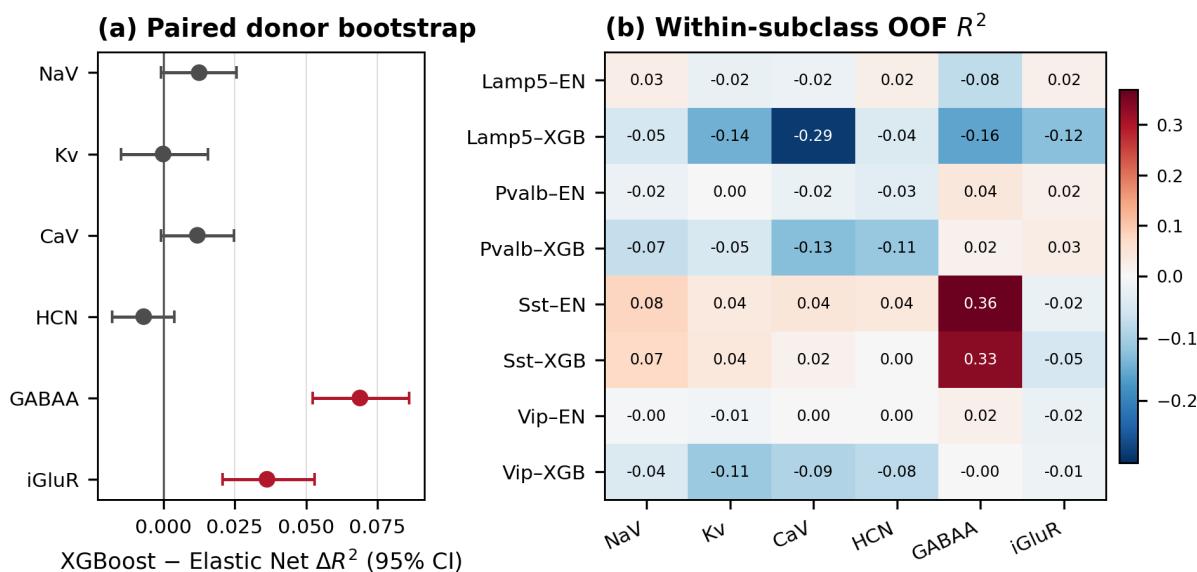
The subclasses meeting the prespecified eligibility rule were Lamp5 (385 cells/171 donors), Pvalb (722 cells/302 donors), Sst (1,504 cells/393 donors), Vip (655 cells/248 donors). Within-subclass estimates are descriptive and unadjusted; they do not establish equivalence. Table S36 provides all four metrics. Table S35 asks the separate question of whether electrophysiology improves a Ridge model already given subclass labels; its intervals are unadjusted exploratory donor-cluster bootstrap intervals based on 5,000 draws.

Complete within-subclass R^2 grid (Table S36).

Subclass	Model	NaV	Kv	CaV	HCN	GABAA	iGluR
Lamp5	Elastic Net	.025	-.022	-.017	.022	-.076	.018
Lamp5	XGBoost	-.054	-.143	-.288	-.043	-.157	-.125
Pvalb	Elastic Net	-.018	.0005	-.019	-.034	.042	.019
Pvalb	XGBoost	-.071	-.047	-.128	-.114	.017	.029
Sst	Elastic Net	.079	.035	.045	.038	.362	-.020
Sst	XGBoost	.074	.039	.015	.001	.334	-.051
Vip	Elastic Net	-.003	-.009	.002	.001	.019	-.019
Vip	XGBoost	-.044	-.112	-.091	-.079	-.0005	-.015

Electrophysiology increment beyond subclass-only Ridge (Table S35).

Module	Subclass Ridge	Ephys + subclass	ΔR^2	95% CI
NaV	.102	.136	.034	[.019, .049]
Kv	.042	.065	.023	[.008, .037]
CaV	.104	.114	.010	[-.002, .021]
HCN	.041	.057	.016	[.002, .029]
GABAA	.177	.319	.142	[.118, .166]
iGluR	.327	.340	.014	[.003, .025]

**Figure S3.** Formal model comparisons and within-subclass prediction. Subclass panels are descriptive and unadjusted.**S13 Repeated Grouped CV and Calibration**

The 5 repeated donor-grouped splits characterize sensitivity to the split, not confidence intervals and not repeated-tuning uncertainty. Tables S37–S39 contain the complete repeated-CV metrics, calibration results, and top-decile enrichment results. The display below gives every model–module combination for repeated R^2 and calibration summaries.

Repeated donor-grouped CV, calibration, and top-decile enrichment (Tables S37–S39).

Module	Model	Repeat median	Repeat range	Primary	Slope	Pred./obs. SD	Top-decile ratio
NaV	Elastic Net	.134	[.126, .137]	.121	.963	.361	1.140
NaV	XGBoost	.140	[.134, .144]	.133	.910	.403	1.155
NaV	MLP	.092	[.077, .101]	.082	.668	.495	1.156
Kv	Elastic Net	.052	[.044, .059]	.046	.980	.219	1.037
Kv	XGBoost	.055	[.050, .066]	.046	.737	.311	1.044
Kv	MLP	.0003	[-.015, .011]	-.015	.465	.467	1.044
CaV	Elastic Net	.097	[.083, .103]	.094	.985	.311	1.110
CaV	XGBoost	.109	[.101, .113]	.106	.879	.373	1.125
CaV	MLP	.040	[.029, .043]	.035	.575	.481	1.095
HCN	Elastic Net	.061	[.058, .066]	.064	.927	.273	1.122
HCN	XGBoost	.046	[.039, .052]	.057	.790	.313	1.126
HCN	MLP	.013	[.011, .017]	.011	.533	.402	1.122
GABAA	Elastic Net	.259	[.249, .263]	.260	1.005	.507	1.149
GABAA	XGBoost	.329	[.328, .340]	.329	1.004	.572	1.165

Tables S37–S39 continued

Module	Model	Repeat median	Repeat range	Primary	Slope	Pred./obs. SD	Top-decile ratio
GABAA	MLP	.314	[.306, .316]	.314	.860	.661	1.156
iGluR	Elastic Net	.258	[.246, .262]	.258	.996	.510	1.109
iGluR	XGBoost	.295	[.284, .300]	.295	.949	.573	1.119
iGluR	MLP	.214	[.206, .222]	.206	.744	.651	1.101

S14 Target Correlations

Tables S40–S42 report Spearman correlations for raw, subclass-centered, and technical-adjusted targets. The strongest raw pair was CaV–iGluR ($\rho = .557$); GABAA–iGluR was weakly negative ($\rho = -.089$). Figure S4 shows the raw matrix; the complete centered and adjusted matrices are provided in Tables S41–S42. Supported specificity is therefore not reducible to the strongest target correlation.

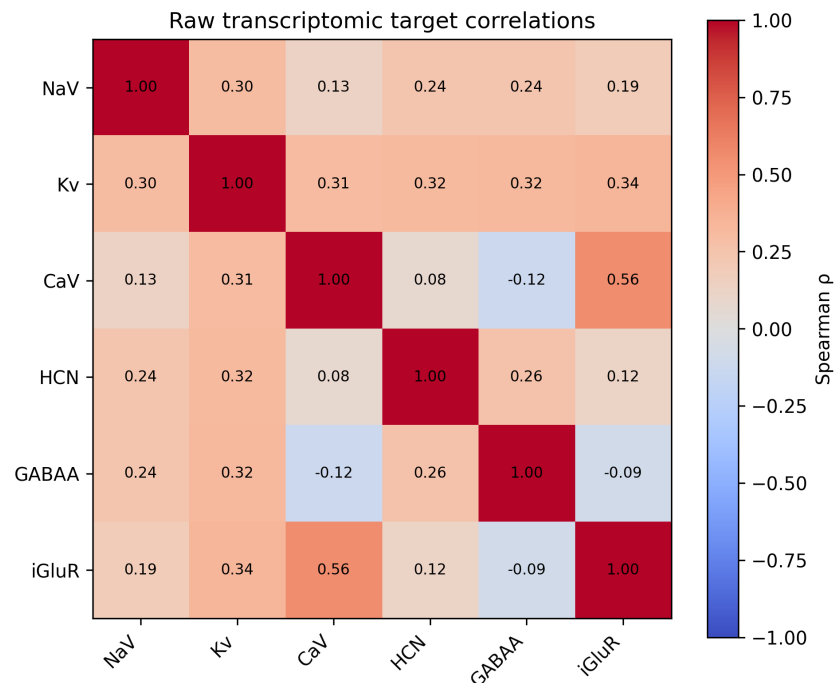


Figure S4. Raw target-correlation matrix (Table S40). Complete subclass-centered and technical-adjusted matrices are reported in Tables S41–S42.

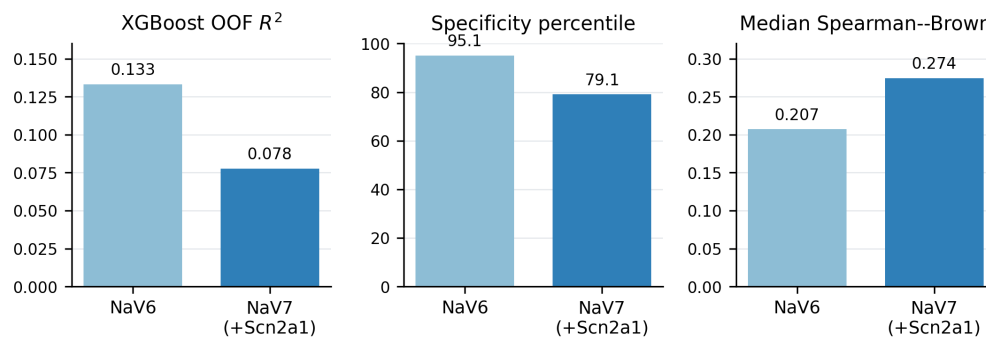


Figure S5. Comparison of the prespecified NaV6 target with the separate NaV7 sensitivity. Values are reported in Table S29. Adding exact *Scn2a1* increased split-half consistency but did not improve OOF prediction or matched-control specificity.

S15 Attribution and Correlated Predictors

The 23 predictors were partitioned into 19 components at $|\rho| > .90$ (Table S43). Held-out TreeSHAP magnitudes were summed within components (Tables S44–S45) and compared with grouped absolute Elastic Net coefficients (Table S46). Because predictors are correlated, these summaries are predictive rather than causal and do not identify unique electrophysiological mechanisms.

Concordance between grouped TreeSHAP importance and grouped absolute Elastic Net coefficients (Table S47).

Module	Groups	Rank ρ	Top-3 overlap	Jaccard	SHAP top 3	Elastic Net top 3
NaV	19	.507	2	.500	C01 C06 C09	C01 C06 C18
Kv	19	.754	2	.500	C01 C14 C17	C01 C09 C17
CaV	19	.533	2	.500	C01 C09 C18	C01 C02 C18
HCN	19	.667	2	.500	C01 C06 C09	C01 C02 C06
GABAA	19	.856	1	.200	C06 C09 C18	C01 C06 C08
iGluR	19	.468	2	.500	C01 C09 C18	C01 C09 C17

Component key for top-three entries (Table S43). C01 = AP downstroke and width (long/short square); C02 = long-square AP fast-trough voltage; C06 = long-square AP threshold voltage; C08 = long-square AP upstroke; C09 = AP upstroke/downstroke ratio (long/short square); C14 = rheobase; C17 = short-square current; C18 = membrane time constant.

S16 Reproducibility and Provenance

All values in this report come from validated artifacts. The accompanying archive contains Tables S1–S50, Figures S1–S5 as PDFs and 300-dpi PNGs, the data-provenance ledger, SHA-256 checksums, software versions, validation results, and reproduction commands. Table S50 links each reported output to its source artifact. The reproducibility package excludes raw Allen data, copyrighted articles, credentials, absolute local paths, and temporary files.

Software environment (Table S48). Python 3.12.3; numpy 2.2.6; pandas 2.2.2; scipy 1.13.0; scikit-learn 1.9.0; xgboost 3.4.0; shap 0.52.0; matplotlib 3.8.4; pyarrow 25.0.0; h5py 3.14.0; joblib 1.5.3; PyYAML 6.0.1; pytest 9.1.1.

Validation and test results (Table S49). *Specificity-control pipeline:* 42 passed in 8.41s; *23-feature modeling pipeline:* 76 passed in 20.43s; *1,000-control extension:* 83 passed in 62.04s (0:01:02); *Curated release:* 107 files; models_fitted=0. Complete evidence paths are supplied in the machine-readable table.

Interpretive boundary. The results support predictive association and, for GABAA and iGluR in the primary family, specificity beyond expression/detection-matched gene sets. They do not measure receptor abundance, conductance, causal regulation, or a formal biological ceiling.